I/B/E/S Earnings Data: Cleaning and Surprise Construction

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> *This report covers 287,844 quarterly earnings announcements (1995-01-03 to 2014-06-30) cleaned from raw I/B/E/S analyst-estimate and CRSP share-price data, and how each announcement's earnings surprise was computed while guarding against look-ahead bias, stock-split artifacts, and ambiguous company identifiers.*

What an "earnings surprise" is

Every quarter, public companies report their actual earnings per share. Wall Street analysts publish estimates of that number beforehand, and the stock market's reaction depends mostly on the GAP between what was reported and what was expected, not the raw number itself. This project calls that gap the "earnings surprise." This report covers a cleaned table of 287,844 such quarterly announcements, spanning 1995-01-03 to 2014-06-30, built from raw I/B/E/S (Institutional Brokers' Estimate System) analyst-estimate data and CRSP share-price data.

Why timing matters: point-in-time discipline

Analyst estimates keep getting updated for years after a company reports, as analysts revise their forecasts for later quarters. That creates a trap: it is easy to accidentally compute a company's "surprise" using an estimate that was published AFTER the announcement, which means the estimate could already reflect the real result. Testing whether a market reacted to information it never actually had would make any later finding meaningless. So every estimate used here is checked to have been published strictly before the announcement it is compared against. Of all announcements, 10,916 consensus snapshots (each one an average across many analysts) were found to have been published on or after the announcement date and were excluded for exactly this reason -- an expected, mechanical filter, not a data error. But publishing beforehand is not the same as being fresh: a matched estimate can still be old, with a median gap of 14 days between the estimate and the announcement, running as high as 1,880 days for late or restated results. That gap (`consensus_staleness_days`) is recorded on every row with a matched estimate, not filtered on -- whether to require a fresher estimate for some later analysis is left to a later slice.

Two ways to measure a surprise

This project computes surprise two different ways. The preferred measure, `sue_analyst`, compares the actual result to the average Wall Street analyst estimate for that quarter, scaled by how much analysts disagreed with each other. It needs real analyst coverage to exist. A mechanical fallback, `sue_seasonal`, does not: it compares the result to the same quarter a year earlier, scaled by how volatile that year-over-year change has historically been for that company. Across the full sample, a real analyst estimate was usable for 67.5% of announcements and the mechanical measure was computable for 76.8% (the two overlap, since both can exist for the same announcement). This project prefers the real analyst estimate whenever it exists and only falls back to the mechanical measure when it does not, which is why the mechanical measure ends up actually USED for a smaller 20.4% of announcements, and 12.1% of announcements have neither and are kept in the data with no surprise value rather than being dropped. Coverage is far from constant over time -- Chart 1, at the end of this report, shows which measure ended up being used in each year of the sample.

How coverage shifted, 1995-2014

Chart 1, at the end of this report, shows, for each announcement year, what share of that year's earnings reports got a real analyst-based surprise, what share fell back to the mechanical measure, and what share had neither. Analyst coverage is thinner in the earliest years of the sample, stays roughly flat (with a dip) through the late 1990s and early 2000s, and then rises through the later 2000s and 2010s as more companies gained regular analyst coverage. The very start (1995-1996) and very end (2014) of the window are both visibly thinner than the interior years: the second year sits at the edge of the one-year lookback the mechanical measure needs, and the final year falls just past a deliberate cutoff in how far the underlying analyst-estimate data was pulled -- both are known edge effects of the sample's boundaries, not data-quality problems.

What the surprise measure looks like across all events

Most earnings announcements are close to what was expected -- a surprise near zero -- with progressively fewer large beats or misses the further from zero you go. That shape (a peak near zero, thinning tails) is real, and Chart 2, at the end of this report, shows it clearly for both benchmarks. Of the announcements that have a surprise value at all, 56.7% land on the positive (beat) side -- a well-documented tendency for reported results to land slightly above expectations. But the two benchmarks are on different scales and neither is simply symmetric around zero, so each needs its own numbers rather than one pooled figure: against analyst expectations, the median surprise is +0.50, and the most extreme 0.1% of announcements reach as low as -68.6 on the downside versus only 36.9 on the upside -- the negative tail reaches further from zero than the positive one, though the two tails carry equal probability mass by construction (each is a 0.1% cutoff). Against the mechanical seasonal benchmark, the median surprise is +0.21, with its own most-extreme 0.1% reaching -25.7 on the downside and 18.5 on the upside -- a similar shape, on a different scale. This combined measure, `sue_primary`, is also not uniformly benchmarked against what analysts expected: 23.2% of the

announcements with a value (58,815 of 253,131) are instead benchmarked against that same company's own result a year earlier (the mechanical fallback measure), not against any analyst estimate.

Data-quality handling: which company, and on what basis

Two separate identification problems had to be handled honestly rather than guessed, and one shades into the other. First, a small share of announcements could not be tied to a single, unambiguous company identifier: 7.5% of announcements carry more than one candidate company ID, most often because the company was reassigned a new internal ID after a corporate restructuring. These are flagged in the data rather than silently resolved one way or another. 2,132 of these are specifically cases where the competing candidate company IDs disagree about whether a stock split happened at all -- so those rows' surprise values are blank for the identifier reason, not because of any split-timing question. Second, stock splits change how many shares a company's earnings are divided by, which can make a perfectly normal result look like a huge surprise if the reported figure and the analyst estimate are not on the same per-share basis. Only 0.6% of announcements (1,647) actually had a stock split land close enough to the announcement to raise this question at all. Of those, 689 were confirmed correct using the company's own prior-quarter figure -- a check measured about 99.13% accurate against Compustat per this project's underlying data notes -- while the remaining 958 are left undetermined because the split-timing evidence was too weak to confirm either way, so the surprise measure is left blank for those rows rather than guessed. Separately, a further 1.8% of announcements (5,195) never had split timing checked at all, for reasons unrelated to split timing itself: 3,063 had no CRSP price data at all to even check, so the pipeline assumes no split adjustment was needed for these (a disclosed tradeoff that is sometimes wrong, and 939 of them still carry a surprise value computed under that assumption), and 2,132 are the identifier-ambiguous rows already discussed above, where competing candidate company IDs disagree about whether a split happened at all. Put together, 2.4% of announcements have some kind of per-share-basis question, of one sort or another; Chart 3, at the end of this report, shows the full breakdown.

What this means for the project

This slice turns raw analyst-estimate and earnings-actuals data into one clean number per company per quarter that says how much that quarter's result beat or missed expectations, with three specific traps -- using information not yet public, mismatched company identifiers, and stock-split artifacts -- checked and either corrected or honestly flagged rather than silently ignored. That is the input the rest of this project needs: later phases will match each of these announcements to the company's actual stock-price behavior in the days and weeks that follow, to test whether the market keeps drifting in the direction of the surprise instead of reacting all at once.

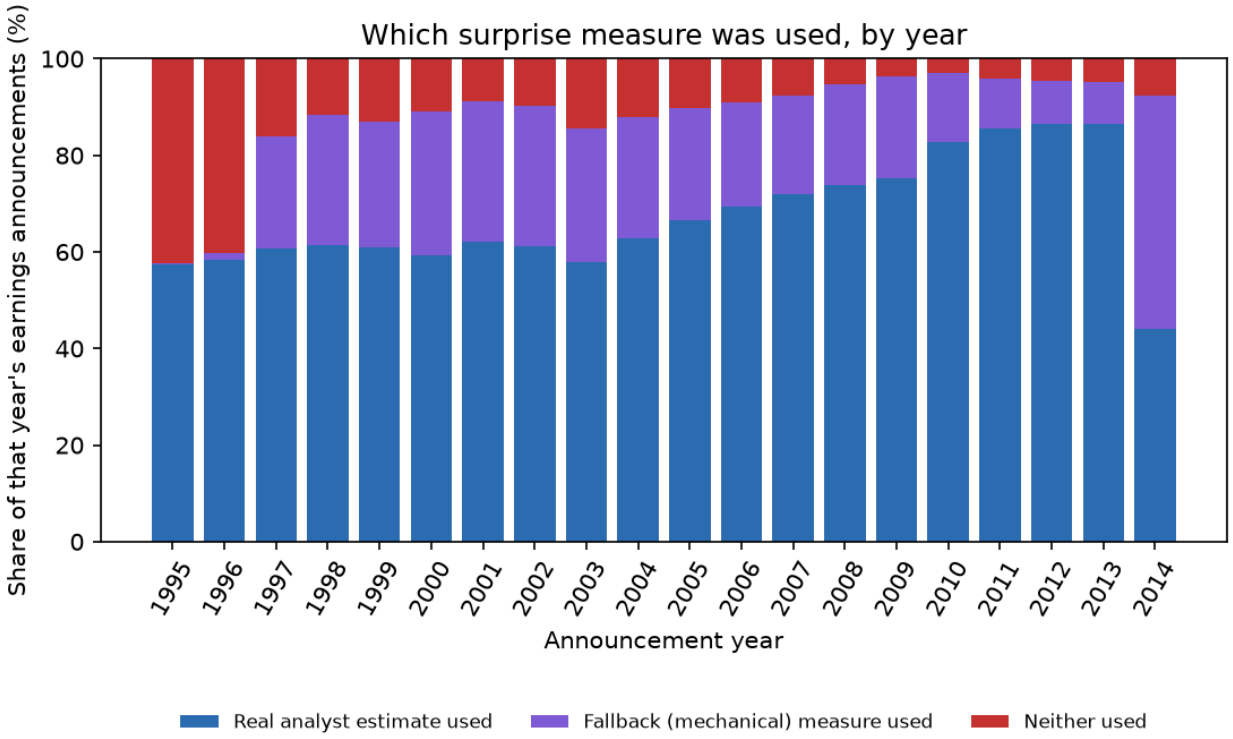


Chart 1 -- Real analyst-based coverage (blue) starts at 57.5% of announcements in 1995, peaks at 86.6% in 2013, and ends the sample at 44.2% in 2014 (see the pull-cutoff edge effect noted above for that final year). The mechanical fallback (purple) fills much of the remaining gap in most years. The share with neither measure (red) falls from 42.4% in 1995 to a low of 3.0% in 2010, though not without some up-years along the way.

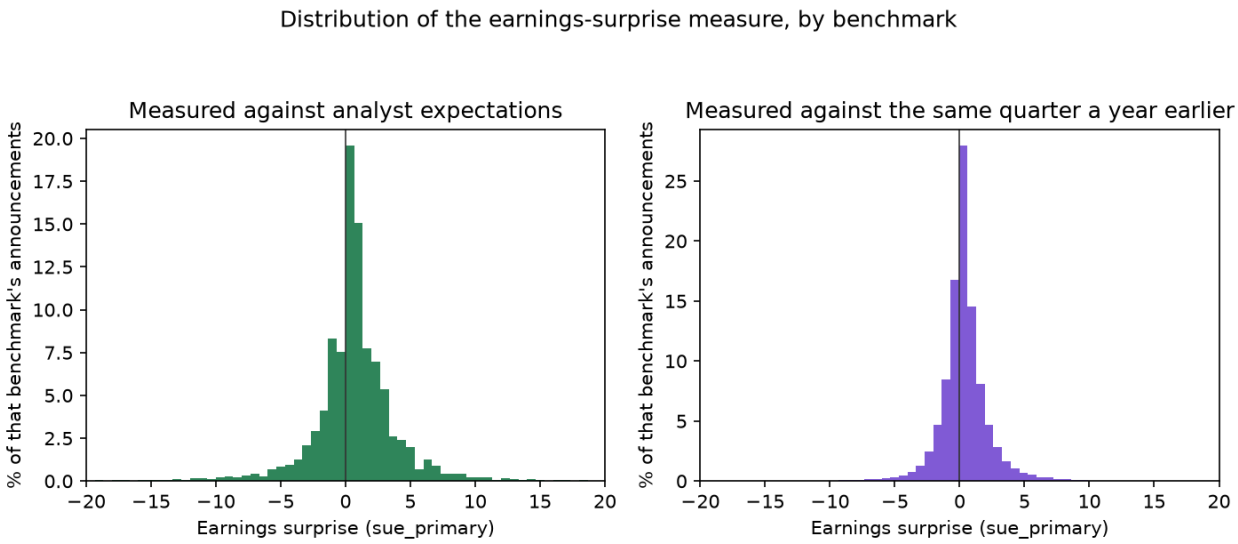


Chart 2 -- Distribution of the earnings-surprise measure (`sue_primary`) across 253,131 announcements that have one, split by which benchmark produced it: analyst expectations (left, 194,316 announcements) or the same quarter a year earlier (right, the mechanical fallback, 58,815 announcements). Each panel's bars show that panel's own share of ITS OWN announcements (not a raw count), so the two panels are visually comparable despite the very different sample sizes. Values are clipped to -20 to 20 for readability; 1.0% of the analyst-benchmarked panel and 0.2% of the seasonal-benchmarked panel have a more extreme surprise than that and are not shown here, but are still included in the underlying data. The two measures are on different scales (analyst disagreement vs. a company's own year-over-year volatility) and should not be pooled for anything scale-sensitive.

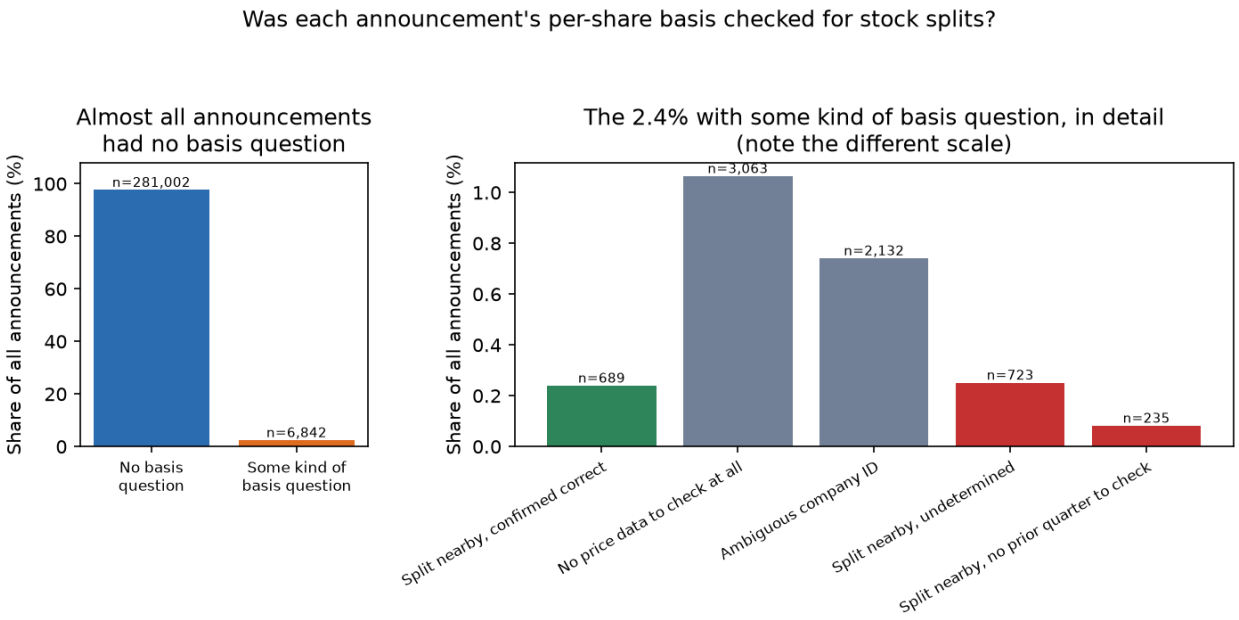


Chart 3 -- Left: 97.6% of all announcements had no per-share-basis question at all; the remaining 2.4% had some kind of basis question (right, note the different scale). Of that 2.4%, only 0.6% (1,647 announcements) actually had a stock split land close enough to the announcement to raise a real split-timing question: 689 were confirmed correct using the company's own prior-quarter figure (a check measured about 99.13% accurate against Compustat), and 958 are left undetermined because the split-timing evidence was too weak either way. The two grey bars are a different, unrelated kind of gap, where split timing was never checked at all: 2,132 announcements where competing candidate company IDs disagree about whether a split happened at all (an identifier problem, not a split-timing one), and 3,063 with no CRSP price data to check at all, for which the pipeline assumes no split adjustment was needed -- a disclosed tradeoff that is sometimes wrong -- and 939 of those still carry a surprise value computed under that assumption. Together, those two never-checked-at-all categories account for 1.8% of all announcements.